

Core Tools Training for Automotive Quality – 2 Days

Duration: 2 Day

Mode: Online / Classroom / On-Site

Certification: QLSS Certificate of Completion



Course Introduction

This intensive 2-day Core Tools training provides in-depth coverage of the six essential tools mandated by IATF 16949:2016: APQP, PPAP, FMEA, MSA, SPC, and the Control Plan. These tools form the foundation of automotive product and process quality assurance, helping organizations ensure consistent product conformity and customer satisfaction.

The training focuses on practical application using real-world manufacturing examples, CSR alignment, and interactive group exercises. Each participant will leave with actionable knowledge and a deep understanding of tool integration.



Pre-requisites

- Basic understanding of quality or manufacturing principles
- Exposure to shopfloor, production, or QA processes
- No prior Core Tools certification required



Learning Objectives

- Understand the structure and application of all six Core Tools
- Align APQP deliverables with customer milestones
- Prepare and interpret PPAP submissions
- Use AIAG-VDA FMEA methods to assess design and process risk
- Create and validate Process Control Plans based on critical characteristics
- Perform MSA studies for gage accuracy, precision, and repeatability
- Apply SPC methods for monitoring process stability and capability

- Integrate all tools into a cohesive quality management workflow
- Map Core Tools usage to IATF 16949 clauses and CSR requirements



Course Agenda



Day 1 – Planning, Risk & Approval Tools

Advanced Product Quality Planning (APQP)

- Phases and timelines
- Linkage with project milestones and team responsibilities
- Deliverables and review checkpoints

2. Control Plan Development

- Types: Prototype, Pre-launch, Production
- Creating a control plan using PFMEA inputs
- Identifying special characteristics
- Linking to reaction plans and work instructions
- Control Plan audit expectations

3. Failure Mode and Effects Analysis (FMEA)

- AIAG-VDA 7-Step Model
- Functional and structural analysis
- SOD ratings and Action Prioritization
- Integration with Process Flow and Control Plan

4. Production Part Approval Process (PPAP)

- PPAP elements and documentation
- Level submission matrix
- CSR alignment and submission status

Group Exercises:

- Map PFMEA to Control Plan
- Create APQP activity list
- Review PPAP checklist and evidence samples



Day 2 – Measurement & Monitoring Tools

5. Measurement System Analysis (MSA)

- Core concepts: Accuracy, Bias, Stability, GRR
- Gage R&R (crossed/nested), attribute MSA
- Acceptance criteria and decision making
- Gage selection and maintenance guidance

6. Statistical Process Control (SPC)

- X-bar R, p, np, c, and u charts
- Cp, Cpk, Pp, Ppk – what they mean and how to use them
- Control limits vs. specification limits
- Interpreting trends and process behavior

Day 2 Activities:

- MSA worksheet for GRR
- SPC chart plotting and Cp/Cpk calculation
- Review of integrated Core Tool workbook



What You'll Receive

- QLSS Certificate of Completion
- PDF Training Handout / Workbook
- Editable templates: PFMEA, Control Plan, GRR, SPC chart
- Individual evaluation and feedback sheet



Who Should Attend

- Quality Engineers, SQA and QA Leads
- Production and Process Engineers
- Manufacturing Supervisors
- Internal Auditors and Core Team Members
- NPD and Supplier Development professionals



Trainer Profile

Delivered by Core Tools practitioners and certified IATF implementation experts with:

- Hands-on experience facilitating PPAP and FMEA reviews
- Tier 1 supplier exposure to Control Plan audits and tool validation
- Strong grasp of CSR expectations and audit nonconformities
- Deep involvement in Core Tools deployment during new product development



Instructional Methodology

- Instructor-led walkthroughs with live tool templates
- Real-world case study examples
- Clause-to-tool mapping from IATF 16949:2016
- Step-by-step tool completion exercises
- Interactive problem-solving using example manufacturing data



What Makes QLSS TISAX Awareness Training Unique

- Full integration of APQP, FMEA, Control Plan, PPAP, MSA, and SPC
- Hands-on templates and worksheets, not just slides
- Strong audit-oriented view: how tools are validated in practice
- Real case examples of risk identification and control measures
- Coverage of latest AIAG-VDA FMEA method in full depth